

SENG1050

Lab 8

In this lab, you will familiarize yourselves with JavaScript by writing a computation script, performing form validation, handling cookies, and working with the Document Object Model (DOM).

[Task 1](#) - [Task 2*](#) - [Task 3*](#) - [Task 4*](#)

Task 1 - Create a Simple Calculator

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>A Very Simple Calculator</title>
5     <meta charset="utf-8">
6     <script type="text/javascript">
7       function calculator() {
8         var x = document.the_form.field_one.value;
9         var y = document.the_form.field_two.value;
10        var p = x * y;
11        alert(x + " times " + y + " is " + p);
12      } // calculator()
13    </script>
14  </head>
15
16  <body>
17    <p>This is a simple calculator. You can add addition, subtraction and division.</p>
18    <form name="the_form">
19      <label for="field_one">Number 1: </label>
20      <input type="text" name="field_one" id="field_one"> <br>
21      <label for="field_two">Number 2: </label>
22      <input type="text" name="field_two" id="field_two"> <br>
23      <input type="button" value="multiply them!" onclick="calculator()">
24    </form>
25  </body>
26 </html>
```

Download the provided *calculator.html* from Canvas. As shown in the image above, the calculator enables the MULTIPLICATION function using Javascript. Please modify the code above by including ADDITION, SUBTRACTION, and DIVISION functions.

You can validate your JavaScript code using a [JavaScript validator](#).

Task 2 - Write a Form Validation Script

Much time and bandwidth can be saved by performing simple checks on a form at the client side before the form is transmitted.

Validation practice

Please enter your name

Please enter your birth year

Going to be a ghost for halloween?

☐ Yes ☐ No ☐ It's a secret

Which of the following sports do you prefer?

- ☐ Football
☐ Soccer
☐ Cricket
☐ Athletics

Your favorite cartoon character:

Submit

Reset

Download the provided *form-validation.html* from Canvas and attempt the following.

1. **Calculate the length of a text string.** From the lecture notes, make use of `document.getElementById("id").value.length`. Use `document.write()` to print out the length of the user's name as entered in the form.
2. Write a JavaScript expression to **validate that one and only one of the checkboxes was selected** from `easyform` (name of the form provided). As a guide, use the following logic. Let A be the event for "no boxes are checked at all" and B be the event for "more than 1 box is checked". If (A OR B), then it is not the case that one and only one box was checked, thus we issue an alert and return false. Otherwise, we thank the user and return true.

3. **Check that a textbox contains only numerical data.** Add a new text input to the form and write a JavaScript function to check that the input is numerical. In your function, you can make use of the following variable which allows a regular expression (regex) check:

```
var pattern = /^-?\d+(\.\d+)?$/;
```

The interpretation is;

`^-?` At the start of the string you may have a minus sign

`\d+` Then you will have one or more digits

`(\.\d+)?` Then, optionally, a decimal point followed by one or more digits

`$` Then the string ends.

To test, use `var a = pattern.test(document.getElementById("id").value)` and check to see if `a` contains "true" or "false".

4. **Calculate the position of a character within a string.** JavaScript's `indexOf` allows us to determine the position of a character (or even a substring) within a string of text. For example,

```
var a = "1 1 was a racehorse, 2 2 was 1 2. 1 1 1 1 race 1 day, 2 2 1 1 2.";
```

```
var b = a.indexOf("w");
```

The value of the variable `b` is now 4 (remember JavaScript begins counting from 0).

Choose a character from the textbox with identity "id" above and use `indexOf` to determine its position. If the character cannot be found, `indexOf` will return the value -1.

Note that these exercises can be used as a starting point for the validation task in your assignment.

Task 3 - Working with Cookies (Challenging - May refer to sample solution)

1. Using the same form from Task 2, write a JavaScript function that will run when the user attempts to submit the form.
This function should write the data to a cookie on the user's computer (use the `document.cookie` property for this).
 2. Write another JavaScript function that will run when the page loads, that will check for the cookie and, if found, pre-fill the form with all of the data from the cookie.
Hint: Read the cookie output first to help structure an autofill strategy. The `split()` function will be helpful here.
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Task 4 - Working with the DOM (Challenging - May refer to sample solution)

As you may have noticed from Task 2, using `document.write` wipes the entire page, leaving only what you specifically passed to the `document.write` method. This is rarely the desired result. Instead, we generally want to work with the elements that already exist on the page, or create our own in a particular position on the page.

- Before an element can be added, it must first be created. Use `document.createElement("tagname")` to create a new element (make sure to assign it to a variable!)
 - You can then position the new element in the DOM by selecting the element that it should be positioned in relation to and using the `insertAdjacentElement` method. For example, to insert a new text input after the reset button the following code could be used:

```
var newInput = document.createElement("input");
newInput.type = "text";
newInput.name = "newtext";
document.easyform.insertAdjacentElement("beforeend", newInput);
```
 - Each element in the DOM has a `remove()` method that will remove it from the DOM. For example, to remove the text input we created next to the reset button, we could use:

```
document.easyform.newtext.remove();
```
1. Create a new text input and button on the form that, when the button is clicked, will add new options for the 'sport' question. The new option should be a radio button, using the text in the new text input.
 2. Write JavaScript code that runs when the page is loaded, and puts a small button next to each radio button for the 'sport' question. When the button is clicked, the corresponding option and radio button should be removed.
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